PONMATHISH T

DAY - 04

10 Marks

1. What is a Network Protocol?

A **network protocol** is a set of rules and standards that devices use to communicate and exchange data over a network.

Main points:

1. It defines how data is formatted and transmitted.
2. It determines how devices establish communication.
3. It controls how data is sent and received.
4. It helps devices understand each other's messages.
5. It can provide error detection and correction.
6. It may provide authentication and security.
7. Different protocols are used for different purposes.
8. Examples include **TCP, IP, HTTP, HTTPS, FTP, DNS, and SMTP**.
9. Protocols ensure reliable and organized communication.
10. Without network protocols, different devices would not be able to communicate properly.

Example: HTTP is used for communication between a web browser and a web server.

2. What is the OSI Model?

The **OSI (Open Systems Interconnection) Model** is a conceptual framework used to understand how data is transmitted between devices over a network.

Main points:

1. It was developed by the **International Organization for Standardization (ISO)**.
2. The OSI model has **seven layers**.
3. Each layer performs a specific networking function.
4. The layers work together to provide communication between devices.
5. It helps network engineers understand how networking works.
6. It makes network troubleshooting easier.
7. It provides a standard way to describe network functions.

8. Data moves through the layers when being transmitted.
9. The receiving device processes the data through the layers in reverse order.
10. The seven layers are **Application, Presentation, Session, Transport, Network, Data Link, and Physical**.

3. Name all Seven OSI Layers in Correct Order

The **seven OSI layers**, from **Layer 7 to Layer 1**, are:

Layer	Name	Main Function
7	Application	Provides network services to applications
6	Presentation	Data formatting, encryption, compression
5	Session	Establishes and manages sessions
4	Transport	Reliable data delivery and flow control
3	Network	Routing and logical addressing
2	Data Link	Frames, MAC addresses, error detection
1	Physical	Transmits raw bits through the medium

Easy way to remember:

A P S T N D P

Application → Presentation → Session → Transport → Network → Data Link → Physical.

4. What is the Purpose of the Application Layer?

The **Application Layer (Layer 7)** is the top layer of the OSI model. It provides network services directly to applications used by users.

Main functions:

1. Provides network services to applications.
2. Allows users to access network resources.
3. Supports web browsing and email services.
4. Handles communication between applications and network services.

5. Provides services such as file transfer and remote access.
6. Supports protocols such as **HTTP, HTTPS, FTP, DNS, and SMTP**.
7. Helps applications communicate with servers.
8. It does not mean the application itself; it provides networking services to applications.
9. It is the closest OSI layer to the end user.
10. It enables users to use network-based applications effectively.

Example: When you open a website, **HTTP/HTTPS** works at the Application Layer.

5. What is the Purpose of the Transport Layer?

The **Transport Layer (Layer 4)** provides **end-to-end communication** between applications running on different devices.

Main functions:

1. Provides end-to-end data delivery.
2. Breaks large data into smaller segments.
3. Reassembles segments at the destination.
4. Provides reliable data delivery when required.
5. Performs error detection and recovery.
6. Controls the flow of data between devices.
7. Uses **port numbers** to identify applications.
8. **TCP** provides reliable, connection-oriented communication.
9. **UDP** provides faster, connectionless communication with less overhead.
10. Helps ensure that data reaches the correct application.

Example: TCP is commonly used for reliable web communication such as HTTPS.

6. What is the Purpose of the Network Layer?

The **Network Layer (Layer 3)** is responsible for **logical addressing and routing data between different networks**.

Main functions:

1. Provides logical addressing using **IP addresses**.
2. Determines the best path for data.
3. Routes packets between different networks.
4. Uses routers to forward packets.
5. Handles packet delivery from source to destination.
6. Helps connect multiple networks together.
7. Performs packet forwarding.
8. IPv4 and IPv6 operate at this layer.
9. Determines where packets should be sent based on destination IP addresses.
10. Enables communication between devices on different networks.

Example: When your computer accesses a website on another network, the **Network Layer uses IP addressing and routing** to deliver the packets to the destination.

7. What is the Purpose of the Data Link Layer?

The **Data Link Layer (Layer 2)** is responsible for reliable communication between devices on the **same local network**.

Main functions:

1. Transfers data between directly connected devices.
2. Converts packets into **frames**.
3. Uses **MAC addresses** to identify devices.
4. Provides error detection using methods such as frame checks.
5. Controls access to the network medium.
6. Helps control the flow of data between connected devices.
7. Works closely with network switches.
8. Divides the layer into **LLC (Logical Link Control)** and **MAC (Media Access Control)** sublayers.
9. Helps ensure frames are delivered to the correct device on the local network.
10. Common technologies include **Ethernet and Wi-Fi**.

Example: A switch uses MAC addresses to forward an Ethernet frame to the correct computer.

8. What is the Purpose of the Physical Layer?

The **Physical Layer (Layer 1)** is the lowest layer of the OSI model. It is responsible for transmitting **raw bits (0s and 1s)** through a physical or wireless medium.

Main functions:

1. Transmits raw bits between devices.
2. Converts bits into electrical, optical, or radio signals.
3. Defines physical cables and connectors.
4. Specifies transmission speeds.
5. Defines signal characteristics.
6. Supports wired and wireless communication.
7. Deals with physical network equipment and media.
8. Includes technologies such as copper Ethernet and fiber-optic cables.
9. Defines aspects such as connectors, cable types, and signal timing.
10. Provides the physical connection required for network communication.

Example: An Ethernet cable carries electrical signals representing bits between a computer and a switch.

9. What is the TCP/IP Model?

The **TCP/IP Model** is a networking model used to describe how devices communicate over networks, including the Internet. It is based on the **TCP/IP protocol suite**.

The commonly used TCP/IP model has **four layers**:

Layer	**Main Function**	**Examples**
4. Application	Provides network services to applications	HTTP, HTTPS, DNS, FTP, SMTP
3. Transport	Provides end-to-end communication	TCP, UDP

Layer	Main Function	Examples
2. Internet	Provides addressing and routing	IP, ICMP
1. Network Access	Handles local network and physical transmission	Ethernet, Wi-Fi

Main points:

1. TCP/IP stands for **Transmission Control Protocol/Internet Protocol**.
2. It is the fundamental protocol suite used on the Internet.
3. **TCP** provides reliable transport when required.
4. **IP** provides addressing and routing.
5. It allows different types of devices and networks to communicate.
6. It is practical and widely implemented.
7. It supports communication across local and global networks.
8. The TCP/IP model has fewer layers than the OSI model.
9. The OSI model has **7 layers**, while the commonly used TCP/IP model has **4 layers**.
10. Together, TCP/IP protocols enable reliable and efficient Internet communication.

10. Explain the Difference Between OSI and TCP/IP Models

The **OSI model** and **TCP/IP model** are networking models used to explain how data is transmitted between devices. However, they differ in their structure and purpose.

Feature	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Number of layers	**7 layers**	**4 layers**
Developed by	ISO	DoD/DARPA
Purpose	Reference/conceptual model	Practical networking model

Feature	OSI Model	TCP/IP Model
Application	Application, Presentation, Session are separate	Combined into Application layer
Lower layers	Data Link and Physical are separate	Combined as Network Access
Transport	Transport Layer	Transport Layer
Network	Network Layer	Internet Layer
Usage	Mainly for learning, design, and troubleshooting	Widely used in real-world networks and Internet
Protocol dependency	Protocol-independent reference model	Based on the TCP/IP protocol suite

OSI Layers:
Application → Presentation → Session → Transport → Network → Data Link → Physical

TCP/IP Layers:
Application → Transport → Internet → Network Access

Conclusion: The OSI model provides a detailed **7-layer reference framework**, while the TCP/IP model is a practical **4-layer model** used for Internet communication.

11. What is Encapsulation?

Encapsulation is the process of **adding control information, such as headers and sometimes trailers, to data as it moves down through the network layers before transmission**.

Process:

1. Data is created by an application.
2. At the **Application Layer**, application data is prepared for communication.
3. At the **Transport Layer**, a TCP or UDP header is added, creating a **segment** (TCP) or **datagram** (UDP).
4. At the **Network Layer**, an IP header is added, creating a **packet**.
5. At the **Data Link Layer**, a frame header and trailer are added, creating a **frame**.
6. At the **Physical Layer**, the frame is converted into **bits/signals**.

7. These bits are transmitted through the network.
8. Each layer adds information needed for communication.
9. The added information helps with addressing, delivery, error detection, and other networking functions.
10. This entire process is called **encapsulation**.

Simple flow:

Data → Segment → Packet → Frame → Bits

12. What is Decapsulation?

Decapsulation is the reverse process of encapsulation. It occurs when the receiving device **removes the headers and trailers added by the networking layers as the data moves upward through the layers**.

Process:

1. The receiving device receives bits from the network.
2. The Physical Layer converts the signals into data.
3. The Data Link Layer processes the frame and removes its header/trailer.
4. The Network Layer processes the IP packet and removes the IP header.
5. The Transport Layer processes the TCP/UDP information.
6. The original application data is recovered.
7. The data is passed to the appropriate application.
8. Each layer performs its required processing.
9. Headers and trailers are removed at the appropriate layers.
10. This complete reverse process is called **decapsulation**.

Simple flow:

Bits → Frame → Packet → Segment → Data

In short:

- **Encapsulation:** Headers/trailers are **added** before transmission.
- **Decapsulation:** Headers/trailers are **removed** at the receiving device.

13. Which Layer Uses IP Addresses?

Answer: Network Layer (Layer 3)

- The **Network Layer** uses **IP addresses**.
- IP addresses identify devices logically on a network.
- They are used for **routing packets** from source to destination.
- Examples: **IPv4** and **IPv6**.
- Routers use destination IP addresses to forward packets.

Example: 192.168.1.10 is an IPv4 address.

14. Which Layer Uses MAC Addresses?

Answer: Data Link Layer (Layer 2)

- The **Data Link Layer** uses **MAC addresses**.
- A MAC address is associated with a network interface.
- It is used for communication within the local network.
- Switches use MAC addresses to forward Ethernet frames.
- MAC addresses are typically represented in hexadecimal format.

Example: 58-E4-EB-41-04-F2 is a MAC address.

15. Which Layer Uses TCP and UDP?

Answer: Transport Layer (Layer 4)

- **TCP (Transmission Control Protocol)** and **UDP (User Datagram Protocol)** operate at the **Transport Layer**.
- TCP provides reliable, connection-oriented communication.
- UDP provides connectionless communication with lower overhead.
- Both use **port numbers** to identify applications.
- Examples:
    - **TCP:** HTTPS, SSH, FTP
    - **UDP:** DNS, DHCP, streaming applications

Protocol/Address	OSI Layer
IP Address	Layer 3 – Network
MAC Address	Layer 2 – Data Link
TCP / UDP	Layer 4 – Transport

16. Which Layer Includes HTTP and HTTPS?

Answer: Application Layer (Layer 7)

- **HTTP (HyperText Transfer Protocol)** operates at the **Application Layer**.
- **HTTPS (HTTP Secure)** also operates at the Application Layer.
- HTTP is used to transfer web pages and other web resources.
- HTTPS provides secure HTTP communication using **TLS encryption**.
- Web browsers and web servers commonly use HTTP/HTTPS.
- HTTPS helps protect data from being read or modified during transmission.

Example: When you visit https://example.com, your browser uses **HTTPS** to communicate with the web server.

17. Why Should Cybersecurity Professionals Understand Networking Layers?

Cybersecurity professionals need to understand networking layers because attacks and security controls can operate at different layers of a network.

Main reasons:

1. Helps understand how network communication works.
2. Makes it easier to identify and analyze cyberattacks.
3. Helps troubleshoot network and security problems.
4. Helps identify suspicious IP addresses and MAC addresses.
5. Helps understand protocols such as **TCP, UDP, HTTP, HTTPS, and DNS**.
6. Helps configure firewalls and network security devices.
7. Helps analyze network traffic using tools such as Wireshark.

8. Helps identify attacks such as scanning, spoofing, DoS/DDoS, and session attacks.
9. Helps determine which network layer is affected during an incident.
10. Helps cybersecurity professionals design effective security controls.

Conclusion: Understanding networking layers provides a strong foundation for **network monitoring, threat detection, incident response, and cybersecurity defense**.

18. Explain How Network Data Travels from a Browser to a Server

When a user enters a website address in a browser, data passes through several networking layers before reaching the server.

Step-by-step process:

1. **Application Layer:**
   The browser creates an **HTTP or HTTPS request** for the website.
2. **DNS Resolution:**
   The browser or operating system uses **DNS** to find the server's IP address from the domain name.
3. **Transport Layer:**
   TCP may establish a connection with the server. For HTTPS, modern web connections can also use **QUIC over UDP**. Port numbers identify the appropriate service.
4. **Network Layer:**
   The data is placed into **IP packets** containing the source and destination IP addresses.
5. **Data Link Layer:**
   The IP packet is placed inside a frame containing **MAC addresses** for the local network hop.
6. **Physical Layer:**
   The frame is converted into electrical, optical, or wireless signals and transmitted.
7. **Routers:**
   Routers forward the packets across different networks toward the server based on the destination IP address.
8. **Server receives the data:**
   The server's networking layers process the incoming data.

9. **Decapsulation:**
   Headers and other control information are processed/removed as the data moves upward through the layers.
10. **Server application responds:**
    The web server processes the request and sends a response back to the browser through the network.

Simple Flow

Browser → HTTP/HTTPS → Transport → IP → MAC/Frame → Network → Server → Decapsulation → Web Server

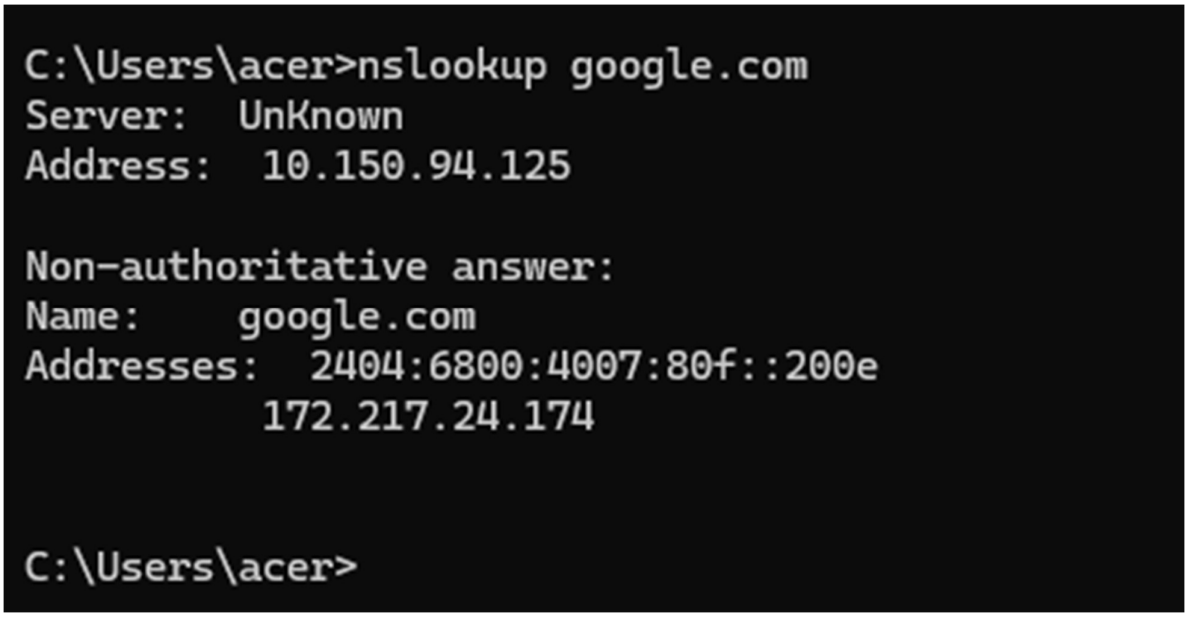
In short: Data is **encapsulated** as it moves down the sender's networking stack and **decapsulated** as it moves up the receiver's stack.

19. Your resolved IP address for google.com (nslookup)

Answer: The resolved IPv4 address of google.com is **172.217.24.174**.
The resolved IPv6 address is **2404:6800:4007:80f::200e**.

20. Number of traceroute hops to google.com

Answer: The traceroute to google.com completed in **12 hops**.



```
C:\Users\acer>tracert google.com

Tracing route to google.com [2404:6800:4007:83e::200e]
over a maximum of 30 hops:

  1     4 ms     4 ms     4 ms  2409:40f4:10f9:23cc::bc
  2     *        *        *     Request timed out.
  3    64 ms    43 ms    52 ms  2405:200:5218:20:3925::ff04
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    55 ms    51 ms    36 ms  2405:200:801:2300::65c
  7     *        *        *     Request timed out.
  8    69 ms    41 ms    58 ms  2001:4860:1:1::17c4
  9    44 ms    72 ms    46 ms  2001:4860:1:1::17c4
 10    69 ms    41 ms    46 ms  2001:4860:0:1::9ee3
 11    67 ms    45 ms    66 ms  2001:4860:0:1::5595
 12    58 ms    30 ms    34 ms  lcmaaa-bc-in-x0e.1e100.net [2404:6800:4007:83e::200e]

Trace complete.

C:\Users\acer>
```

```
C:\Users\acer>ping google.com

Pinging google.com [2404:6800:4007:80f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:80f::200e: time=88ms
Reply from 2404:6800:4007:80f::200e: time=51ms
Reply from 2404:6800:4007:80f::200e: time=41ms
Reply from 2404:6800:4007:80f::200e: time=84ms

Ping statistics for 2404:6800:4007:80f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 88ms, Average = 66ms

C:\Users\acer>
```